

SET UP HYDROLOGIC MODEL

GLY606 Water Data Analysis & Modeling

Nov 11th 2024



Step 1: Prepare meteorological forcing data

- `conda activate xesmf_env`

first step aims to calculate the required variables with correct units as VIC input

- `python 1_prep_met_forcing.from_era5.py <#1_site_id> <#2_era5_met_dir>
<#3_orig_rsln_dir> <#4_start_year> <#5_end_year>`

These scripts are located in Github: [GLY606_2024/final_project/met_prep_python_code](https://github.com/University-at-Buffalo/GLY606_2024/tree/main/final_project/met_prep_python_code)

second step aims to regrid the met forcing at domains for the model

- `python python 2_regrid_met_forcing.py <#1_site_id> <#2_orig_rsln_dir >
<#3_regrid_dir> <#4_start_year> <#5_end_year> <#6_domain_file>`

Step 1: Prepare meteorological forcing data

- `conda activate xesmf_env`

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You need to create **a folder that will store the processed ERA5 forcing at the original resolution**, as well as **a folder that will store the regridded dataset**.

second step aims to regrid the met forcing at domains for the model

- `python python 2_regrid_met_forcing.py <#1_site_id> <#2_orig_rsln_dir >
<#3_regrid_dir> <#4_start_year> <#5_end_year> <#6_domain_file>`

Step 2: Create a model run folder

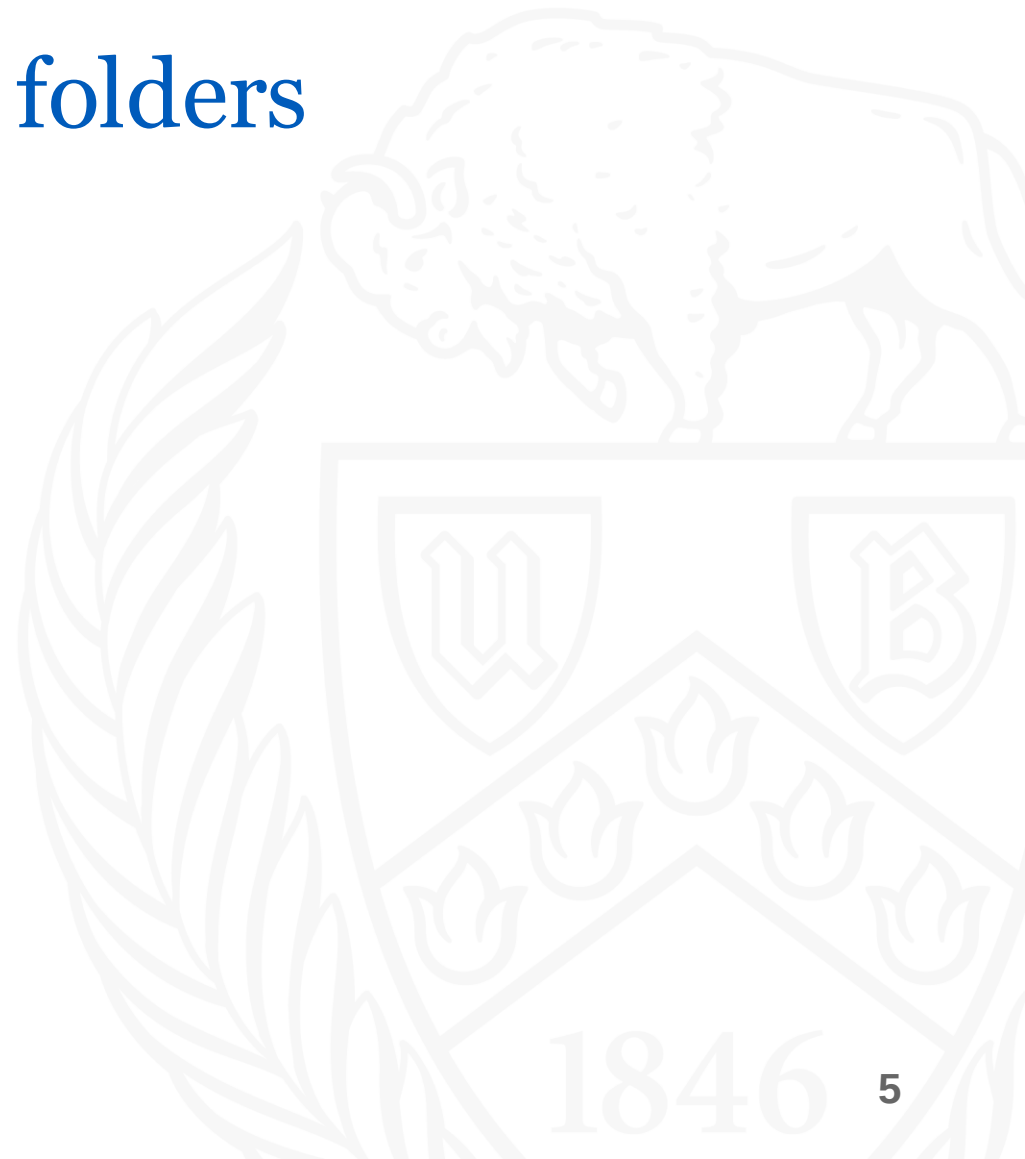
- `cd ~/model`
- `mkdir final_project`
- `cd final_project`
- `mkdir vic mizuroute`
- `cd vic`
- `mkdir config param state output output/log`

Example folder structure

- model
 - final_project
 - vic
 - config
 - param
 - state
 - output
 - log
- mizuroute

Step 3: Move files to the targeted folders

- `~/model/final_project/vic/param`
 - `namerica_domain.<siteid>.nc`
 - `namerica_params.<siteid>.update.nc`
- `~/model/final_project/vic/config`
 - `basin.template.global`



Step 4: Prepare configuration file

- `cd ~/model/final_project/vic/config`
- `cp basin.template.global basin.<siteid>.global`
- `vim basin.<siteid>.global`
 - <STARTYEAR>
 - <ENDYEAR>
 - <SITEID>
 - <FORCING_DIR>

If you followed the folder structure as I suggested, you only need to modify the four types of placeholders on the left.

Otherwise, you will need to make sure that the configuration file contains the correct path to 1) the domain file and 2) the parameter file.

You may need to replace each type of placeholder multiple times.

`basin.template.global` is located in Github: [GLY606_2024/final_project/vic_config_template](#)

Step 5: Run the model

- `cd ~/model/final_project/vic/config`
- `~/model/VIC/vic/drivers/image/vic_image.exe -g basin.<siteid>.global`



Model spin-up

- The land surface model “spin-up” process is used to generate a reasonable initial state from which the model can proceed.
- For example, if we want to conduct a 10-year model spin-up and our official model run to start from **2010-01-01**. How should we set up?

